Experiment No. 4	
CODE CONVERSION	
Course Code: CPE 013	**Program: BSCpE**
Course Title: Logic Circuits and Design	**Date Performed: 01/15/2026**
Section: 22S2	**Date Submitted: 01/20/2026**
Members: Bulaong, Ross Andrew	**Instructor:** Engr. Noel B. Lusung
Florencio, David	
Que, Josey Leroy	
Malabanan, Nadine	
Zapanta, Russell	

1. Objective:

This activity aims to implement a Binary to Gray code converter and vice versa using exclusive-OR gates

2. Intended Learning Outcomes (ILOs):

The students should be able to:
2.1. Understand and analyze the concept of code conversion between Binary and Gray Code.
2.2. Construct truth table and circuit diagram for code conversion of Binary to Gray code and vice versa.

3. Discussion

A Gray code is an encoding of numbers so that adjacent numbers have a single digit differing by 1. The term Gray code is often used to refer to a "reflected" code, or more specifically still, the binary reflected Gray code.

It is named for Bell Labs researcher Frank Gray, who described it in his 1947 patent submittal on Pulse Code Communication. He did not call it a Gray Code, but noted there was no name associated with the novel code and referred to it as a Binary Reflected Code for the way he determined the groupings and number representations. When the patent was granted in 1953 others began to refer to the encoding scheme as the Gray Code.

A Gray Code is not weighted, the columns of bits do not reflect an implicit base weight as the Binary number system does. In the Binary number system, the least significant bit (right-most) column is weighted as 2^0 (1); the second column, 2^1 (2); the third 2^2 (4), and so on, each column representing the base raised to a power. The final value is determined by multiplying the bit by the column weight and adding the column results, so in Binary the 4-bit number '0011' represents 1*2 + 1*1 = 3.

Table 4.1. Decimal, Binary and Gray Code

Decimal	Binary	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111

6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111

Source/Reference: https://mathworld.wolfram.com/GrayCode.html
https://www.allaboutcircuits.com/technical-articles/gray-code-basics/

4. Resources:

Materials:

- Personal Computer
- Internet connection
  Simulation Tools:
  - TinkerCAD
  - Logicly

5. Procedure:

1. Construct the circuit as shown in Figure 4.1. Connect the INPUTS (B3,B2, B1 and B0) to the data switches and the OUTPUT (G3,G2, G1 and G0) to the LED as indicator for Gray Code result.

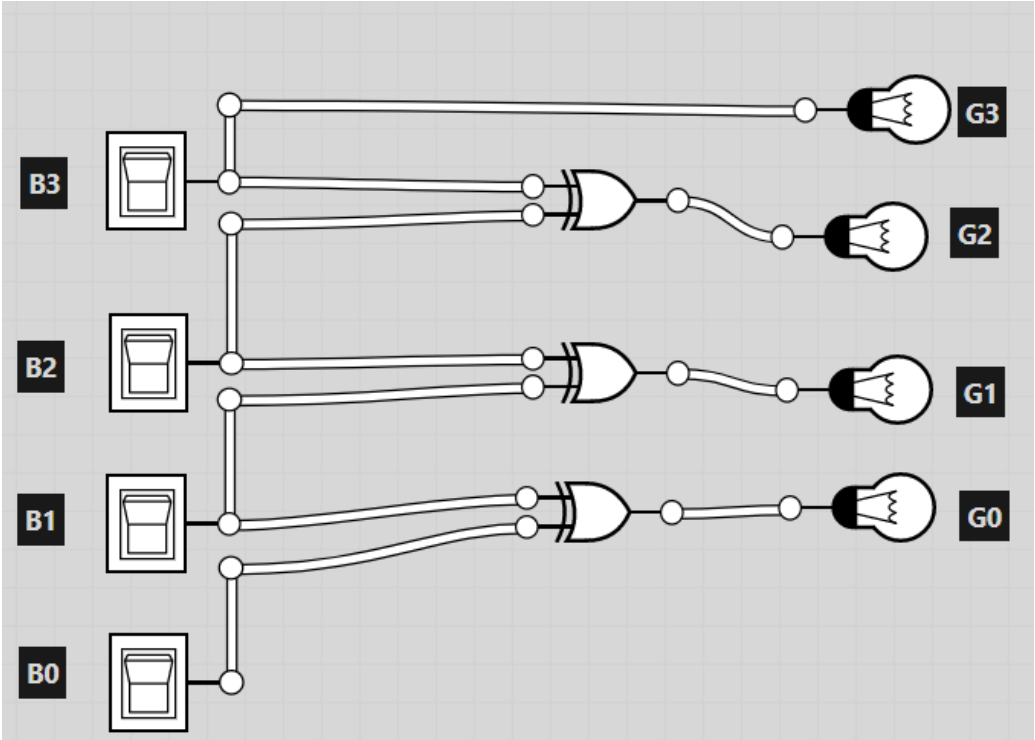
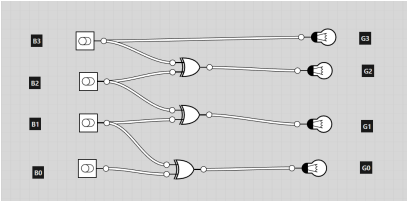
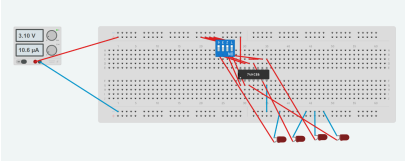
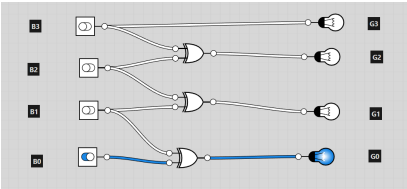
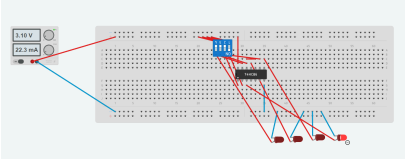
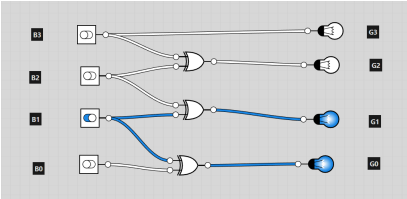
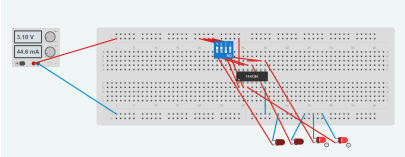
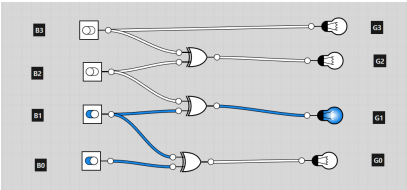
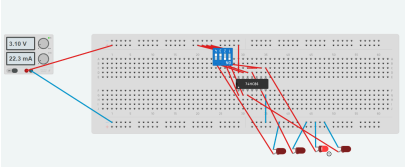
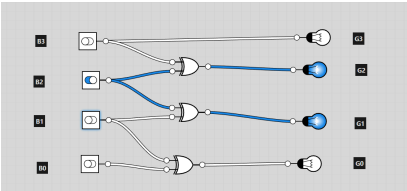
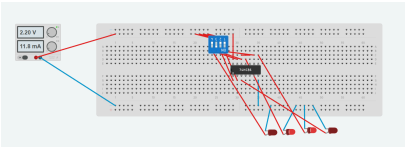


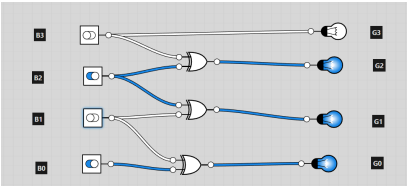
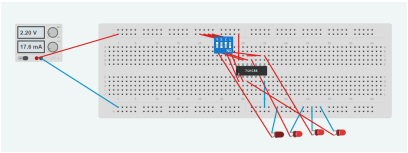
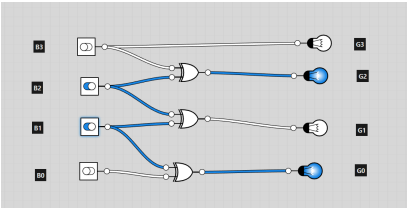
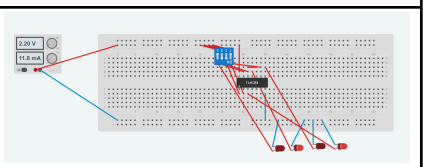
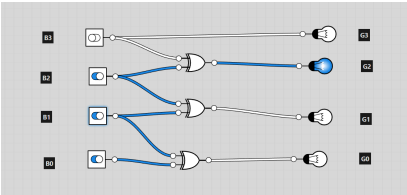
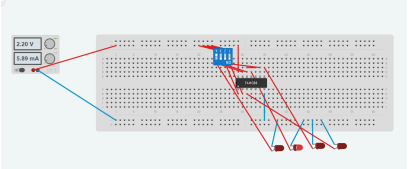
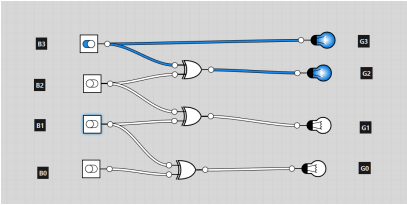
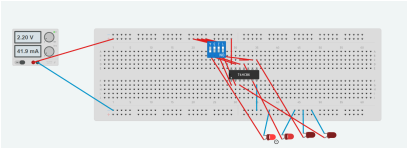
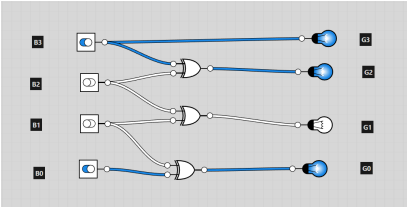
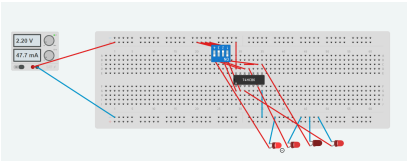
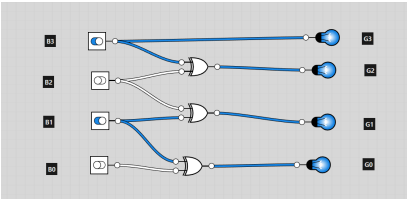
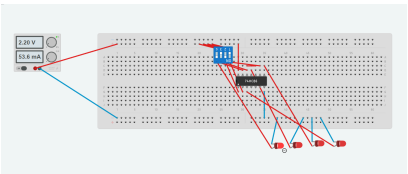
Figure 4.1. Binary to Gray Code Diagram

2. Apply the different combinations to the data switches and observe the corresponding output in

every combination. Record the answers in Table 4.2 by taking screenshot of each combination in Logicly and TinkerCAD.

Table 4.2. Binary to Gray Code Simulation Confirmation

Binary	Gray Code	Logicly	TinkerCAD
0000	0000		
0001	0001		
0010	0011		
0011	0010		
0100	0110		

0101	0111		
0110	0101		
0111	0100		
1000	1100		
1001	1101		
1010	1111		

Link for Tinkercad:
https://www.tinkercad.com/things/3YgMLbJM2JA-code-conversion?sharecode=MNSG_YkiO9SwjQztUVzaO4Txf3Xsxthw30aWIUML-XE

6. Data and Results:

1. Construct the table conversion for Gray code to binary in Table 4.3.

Gray Code	Binary
0000	0000
0001	0001
0011	0010
0010	0011
0110	0100
0111	0101
0101	0110
0100	0111
1100	1000
1101	1001
1111	1010

Table 4.3 Gray Code to Binary

2. Design and construct the circuit diagram for Gray Code to Binary.

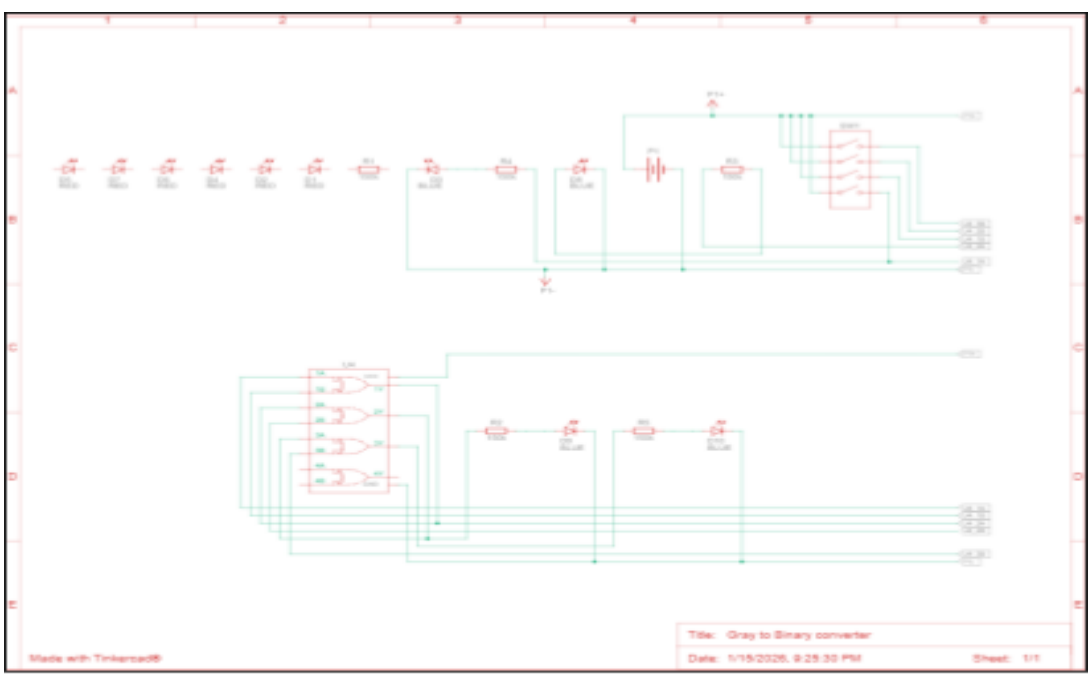
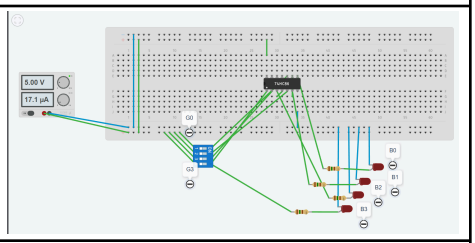
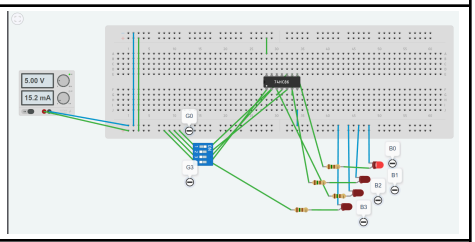
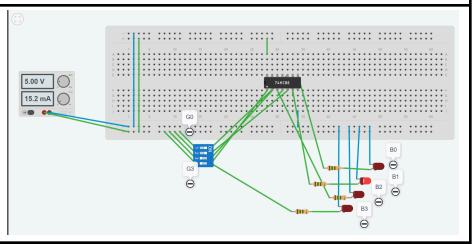
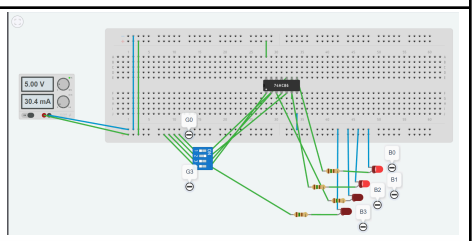
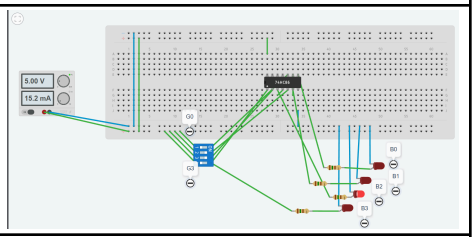
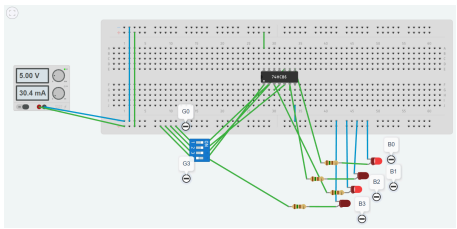
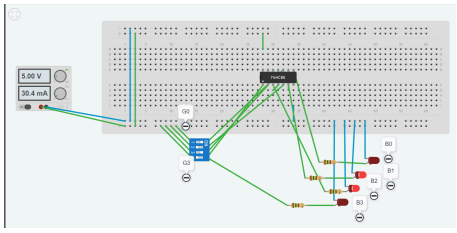
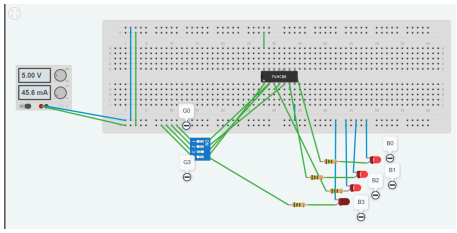
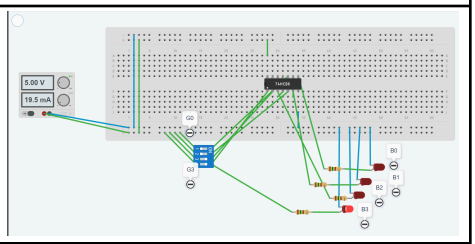
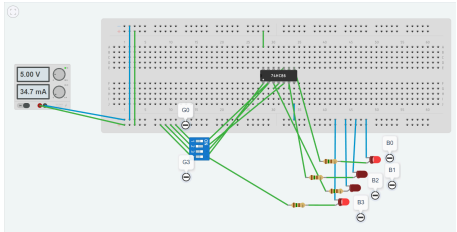
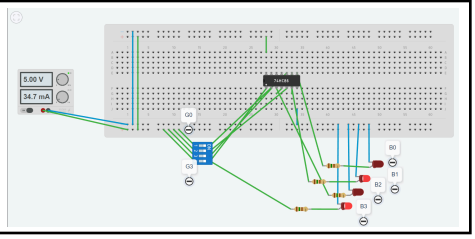


Figure 4.2. Gray Code to Binary Diagram

3. Apply the different combinations to the data switches and observe the corresponding output in every combination. Record the answers in Table 4.4 by taking a screenshot of each combination in Logicly and TinkerCAD.

Table 4.4 Gray Code to Binary Simulation Confirmation

Gray Code	Binary	Logicly	TinkerCAD
0000	0000		
0001	0001		
0011	0010		
0010	0011		
0110	0100		
0111	0101		

0101	0110		
0100	0111		
1100	1000		
1101	1001		
1111	1010		

7. Final Output

Screenshot or print screen of TinkerCAD application different output simulations (5 points deductions for only one output in every required circuit). Also, upload a video of the simulated circuit naming or talking about the shown input combinations. Failure to comply to each instruction will resulted to 5 points deductions. Please include the recorded simulation video in this section and write at least one sentence definition or label for each figure or image

TINKERCAD LINK: **https://www.tinkercad.com/things/bcWQRer3CXe-gray-code-to-binary**

8. Conclusion:

Write a one paragraph (at least 10 sentences) Conclusion for this activity.

Bulaong

I gained a better understanding of Binary and Gray code conversion using XOR gates, which can be helpful when working with digital and electronic systems in the future.

Florencio

In this activity, I learned how to convert Binary and Gray codes using XOR gates. This lesson may help me apply code conversion in real-life digital and electronic systems in the future.

Que

For the tinkercad breadboarding, specifically the gray code to binary simulation, we have discovered the dip x4 switch which made a huge difference on executing the schematics. This discovery made it easier to test the LEDs as it shows its efficiency when doing the wirings.

Malabanan

In this lab, I learned how to convert Binary codes to Gray codes and vice versa using XOR gates. I built the Gray to Binary circuit in TinkerCAD, tested it with the 4-bit DIP switch, and created the schematic diagram. This activity helped me understand code conversion in practice and how digital circuits respond to different inputs.

Zapanta

Following the instructions makes it easier to do this experiment. It has enough instructions on what we should do. I also learned that it is possible to convert from binary to gray code.

9. Questions:

1. What do you think is the importance of knowing code conversion? How does it work in real-life industry applications?

Knowing code conversion is important because different digital systems use different codes to represent data. It allows devices to communicate correctly by translating information from one format to another. In real-life industries, code conversion helps computers, microcontrollers, and digital circuits process data accurately. Without code conversion, systems like sensors, displays, and processors would not work together properly.

2. What other industry related devices do you think use code conversion? Cite at least one example and explain it in at least 2-4 sentences.

   One example of a device that uses code conversion is a digital thermometer. It converts analog temperature signals into digital or binary codes so the microcontroller can process and display the temperature accurately. Another example of code conversion is an elevator control system. It converts floor numbers entered by users into binary codes so the controller can determine which floor to move to and display the correct floor number on the panel.

10. Assessment (Rubric for Laboratory Performance):

Laboratory Performance Rubric (2)

Criteria	Ratings			Pts
Manipulative Skills	**3 pts** **Proficient** Student always demonstrate needed skills.	**2 pts** **Acceptable** Student occasionally demonstrate needed skills.	**1 pts** **Beginner** Student do not demonstrate needed skills.	3 pts
Experimental Set - up	**3 pts** **Proficient** Student are able to set-up the material with minimum supervision.	**2 pts** **Acceptable** Student are able to set-up the materials with supervision.	**1 pts** **Beginner** Student are unable to set-up the materials.	3 pts
Process Skills	**3 pts** **Proficient** Student always demonstrate targeted process skills.	**2 pts** **Acceptable** Student occasionally demonstrate targeted process skills.	**1 pts** **Beginner** Student do not demonstrate targeted process skills.	3 pts
Application of Technical Writing Skills	**3 pts** **Proficient** Correct spelling, grammar, and sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	**2 pts** **Acceptable** Some minor incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	**1 pts** **Beginner** Many incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	3 pts
Time Management / Conduct of Experiment	**3 pts** **Proficient** Student finish ahead of time with complete data and time to revise data.	**2 pts** **Acceptable** Student finish on time with incomplete data.	**1 pts** **Beginner** Student do not finish on time with incomplete data.	3 pts
Complete Requirements of Laboratory Report	**3 pts** **Proficient** Student submitted a complete requirements set by the teacher.	**2 pts** **Acceptable** Student submitted an almost complete Laboratory Report according to requirements set by the teacher.	**1 pts** **Beginner** Student submitted a lot of missing requirements in the Laboratory Report according to requirements set by the teacher.	3 pts
Follow the Correct Format of Laboratory Report	**3 pts** **Proficient** Student submitted a correct format of Laboratory Report according to requirements set by the teacher.	**2 pts** **Acceptable** Student submitted an almost correct format of Laboratory Report according to requirements set by the teacher.	**1 pts** **Beginner** Student did not follow at all the format of Laboratory Report according to requirements set by the teacher.	3 pts
Ability to do independent work	**3 pts** **Proficient** Student do not need to be supervised by the teacher.	**2 pts** **Acceptable** Student require occasional supervision by the teacher.	**1 pts** **Beginner** Student require supervision by the teacher.	3 pts
				Total Points: 24